

Revision & Problems

Sample space: Set of all possible outcomes of an experiment (S, Ω) (Ω)

$$\omega \in S$$

Events: Subset of sample space

$$A \subseteq S \quad \text{Set of all events}$$

$$2^{|S|}$$

Events $\neq$ outcomes

$$\mathcal{F} \subseteq \text{set of all events}$$

subset of S

elements of S

Probability of an event

$$P: \mathcal{F} \rightarrow [0, 1]$$

Axioms of probability:

$$(i) \quad \forall A \subseteq S, P(A) \geq 0$$

$$(ii) \quad P(S) = 1$$

$$(iii) \quad A_i \cap A_j = \emptyset$$

$$P(A_i \cup A_j) = P(A_i) + P(A_j)$$

A_i 's mutually disjoint

$$P(\cup A_i) = \sum P(A_i)$$

When outcomes are "equally likely", then
define $P(A) = \frac{|A|}{|S|}$ & $|S| < \infty$
& $|S| \neq 0$

Independent events:

A, B are independent if
 $P(A \cap B) = P(A) P(B)$

Conditional probability: $P(B|A) \cdot P(A)$
 $P(A|B) = \frac{P(A \cap B)}{P(B)}$ (if $P(B) > 0$)

Independent events: $P(A|B) = P(A)$ or $P(B|A) = P(B)$

Total probability theorem:

$$P(A) = P(A|B)P(B) + P(A|B^c)P(B^c).$$

Generalized total probability theorem:

$B_1, B_2, \dots, B_n$ partition of S

i.e. (i) $B_1 \cup B_2 \cup \dots \cup B_n = S$

(ii) for $i \neq j$, $B_i \cap B_j = \emptyset$

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots \\ \dots + P(A|B_n)P(B_n).$$

Bayes rule:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Generalized Bayes rule:

If $B_1, B_2, \dots, B_n$ is a partition of S ,
then

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A)}$$

$$\Rightarrow P(B_i|A) = \frac{P(A|B_i)P(B_i)}{P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \dots + P(A|B_n)P(B_n)}$$

Random variable

$$X: S \rightarrow \mathbb{R}$$

$$X(\omega) \in \mathbb{R}$$

Distribution of random variable

Discrete r.v. : Range of X is a countable subset of $\mathbb{R}$
 $\{k_1, k_2, \dots\}$

Discrete Probability Distribution

Probability mass function (PMF)

$$p_X(k) = \frac{P(X=k)}{1}$$

$k \in \text{Range of } X$
 $k \notin \text{ " " " "}$

PMF of $X : p_X : \mathbb{R} \rightarrow [0, 1]$

Properties :

$$(i) \sum_{k \in \mathbb{R}} p_X(k) = \sum_{k \in \mathbb{R}} P(X=k) = 1$$

$$(ii) p_X(k) \geq 0$$

$\sum_{k \in \text{Range of } X} p_X(k)$

Cumulative distribution function (CDF)

Define $F_X : \mathbb{R} \rightarrow [0, 1]$

$$F_X(t) = P(X \leq t)$$

Basic properties of CDF

(i) F_X is an increasing funcⁿ on $\mathbb{R}$

$$(ii) \lim_{t \rightarrow +\infty} F(t) = 1 \quad \& \quad \lim_{t \rightarrow -\infty} F(t) = 0$$

(iii) F_X is right continuous

$$\lim_{h \rightarrow 0} F(t+h) = F(t) \quad \forall t \in \mathbb{R}$$

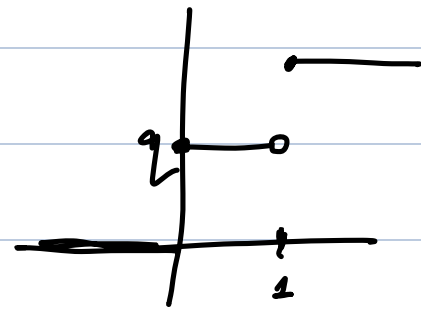
Recover PMF from CDF

$$P(X=a) = P(a-1 < X \leq a) = F_X(a) - F_X(a-1)$$

$F_X(a)$
 $F_X(a-1)$

$\mathbb{R} \rightarrow [0, 1]$

$$F_X(t) = \begin{cases} 1 & t \geq 1 \\ q & t \in [0, 1) \\ 0 & t < 0 \end{cases}$$



$$p_X(k) = \begin{cases} P(X=0) = q & ; k=0 \\ P(X=1) = 1-q = p & ; k=1 \end{cases}$$

$$P(a < x \leq b) = F_X(b) - F_X(a)$$

$$\begin{aligned} F_X(t) &= P(X \leq t) \\ &= \begin{cases} P(X=0) = q & ; 0 \leq t < 1 \\ P(X=0) + P(X=1) = 1 & ; t \geq 1 \end{cases} \\ &\quad \begin{matrix} 0 & t < 0 \end{matrix} \end{aligned}$$

Types of discrete probability distribution:

Bernoulli: $X \sim \text{Bern}(p)$ Range of $X = \{0, 1\}$
 $P(X=0) = q$ & $P(X=1) = 1-p$

$$p_X(k) = \begin{cases} p & k=0 \\ 1-p & k=1 \end{cases}$$

$$F_X(t) = \begin{cases} 1 & t \geq 1 \\ p & t \in [0, 1) \\ 0 & t < 0 \end{cases}$$

Binomial: $X \sim \text{Bin}(n, p)$

$X = \# \text{ successes in } n \text{ trials}$

PMF: $p_X(k) = {}^nC_k p^k (1-p)^{n-k}$, $0 \leq k \leq n$

CDF: $F_X(t) = P(X \leq t)$

Geometric: $X \sim \text{Geom}(p)$, $p \in (0,1]$

$X = \#$ trials until a success in independent

Bernoulli trial

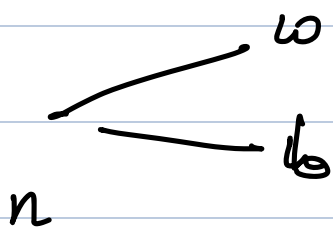
PMF: $P(X=k) = (1-p)^{k-1} p$ $k=1,2,\dots$

CDF: $F_X(t) = P(X \leq t) = 1 - P(X > t)$

$= 1 - \sum_{k > t} P(X=k)$

$= 1 - (1-p)^t$

Hypergeometric:



$X \sim \text{HGeom}(w, b, n)$

PMF of $X = \frac{{}^wC_k \cdot {}^bC_{n-k}}{{}^{w+b}C_n}$; $0 \leq k \leq w$

$E[X] = \frac{wn}{w+b}$

Further topics:

Poisson, Negative Binomial, expectation, variance, joint distribution, marginal distribution, LOTUS, 2D LOTUS, covariance

Problems:

A fair coin is flipped twice. let

$X = \#$ heads in the 2 tosses

$Y =$ indicator r.v for the tosses landing in the same way

(a) Find the joint PMF of X & Y .

(b) Find the marginal PMF of X & Y .

(c) Are X & Y independent?

$$S = \{HH, TH, TT, HT\}$$
$$\text{Range of } X = \{0, 1, 2\}$$
$$\text{" " } Y = \{0, 1\}$$
$$Y = \begin{cases} 1 & \text{if } \{HH, TT\} \\ 0 & \text{o/w} \end{cases}$$

$X \backslash Y$	0	1	
0	0	$1/4$	$1/4$
1	$1/2$	0	$1/2$
2	0	$1/4$	$1/4$

② Mark plays a lottery game. He spends \$2 each week to purchase a lottery ticket for this game.

Suppose the probability of winning a \$4 prize on each ticket is $\frac{1}{30}$, and the probability of winning the jackpot on each ticket is $\frac{1}{3 \times 10^8}$.

i) If Mark purchases a total of 52 tickets in the year 2025.

Find the expected number of times Mark wins a \$4 prize this year?

$$\begin{aligned} n &= 52, \quad p = \frac{1}{30} \\ E[X] &= np \\ &= \frac{52}{30} \end{aligned}$$

(ii) If Mark has purchased 8 tickets since Jan 1, 2025, what is the probability that Mark won the \$4 prize twice since Jan 1, 2025?

$$X \sim \text{Bin}(8, 1/30)$$
$$P(X=2) = \binom{8}{2} \cdot \left(\frac{1}{30}\right)^2 \cdot \left(\frac{29}{30}\right)^6$$

(iii) What is the expected number of tickets needed until a Jackpot is won (suppose he could play the game forever)

$$P(X=k) = (1-p)^{k-1} p$$

$$E[X] = \frac{1}{p} = 3 \times 10^8$$

③ Let X be the geometrically distributed random value, with probability of success p_x ;
 Y be the binomially distributed random variable with parameters n, p_y .
 let X & Y be independent.

(a) Find the probability that $X = Y + 1$?
 (Simplify your answer)

Sol.ⁿ: $X \sim \text{Geo}(p_x)$
 $Y \sim \text{Bin}(n, p_y)$

$X = Y + 1$ is equivalent to the union of events $\{X = k+1, Y = k\}$ for $0 \leq k \leq n$

$$\sum_{k=0}^n P(X = k+1, Y = k)$$

$$= \sum_{k=0}^n P(X = k+1) P(Y = k)$$

disjoint $\begin{cases} X=1, Y=0 \\ X=2, Y=1 \end{cases}$

$$= \sum_{k=0}^n (1-p_x)^k p_x \cdot {}^n C_k \cdot p_y^k (1-p_y)^{n-k} \rightarrow$$

④ If an intercontinental ballistic missile (ICBM) is launched from a nuclear submarine in a certain part of the world, a radar system might be able to detect its presence while airborne and launch a second missile to intercept the ICBM.

Let M : event that ICBM has been launched.

$$P(M) = 0.001$$

D : event that the radar system alerts of the presence of a missile

$$P(D|M) = 0.99$$

$$\& P(D^c|M^c) = 0.99$$

(a) Let E denote the event that the radar system makes a mistake.

Compute $P(E)$

$$\begin{aligned} \rightarrow P(E) &= P(E|M)P(M) + P(E|M^c)P(M^c) \\ &= P(D^c|M)P(M) + P(D|M^c)P(M^c) \end{aligned}$$

$$\rightarrow = 0.01 \times 0.001 + 0.01 \times 0.999$$

(b) Given that the radar system detects the presence of an ICBM, compute the probability that an ICBM has actually been launched.

$$P(M|D)$$

Sol.ⁿ:
$$P(M|D) = \frac{P(D|M)P(M)}{P(D)} \quad (\text{By Bayes' rule})$$

Now, By total probability theorem

$$\begin{aligned} P(D) &= P(D|M)P(M) + P(D|M^c)P(M^c) \\ &= 0.99 \times 0.001 + 0.01 \times 0.999 \end{aligned}$$

So,

$$P(M|D) = \frac{0.99 \times 0.001}{P(D)}$$

$$\lambda = E[X^2] - (E[X])^2$$

$$E[X^2] = \lambda + \lambda^2$$

(5)

Let $X \sim \text{Pois}(\lambda)$. Let $Y = 1 + X$

Compute (a) $E[Y^2]$ $= E[(1+X)^2]$

(b) $E[Y]$

$$\begin{aligned} &E[1] + 2E[X] \\ &+ E[X^2] \end{aligned}$$

Solⁿ: $E[Y^2] = E[(1+X)^2] = E[1+X^2+2X]$
 $= 1 + E[X^2] + 2E[X] \rightarrow \text{①}$

Now $E[X] = \lambda$

2 $E[X^2] - (E[X])^2 = \lambda$

$\Rightarrow E[X^2] = \lambda + \lambda^2$

① reduces to

$E[Y^2] = 1 + \lambda + \lambda^2 + 2\lambda$
 $= \lambda^2 + 3\lambda + 1$

$E[1/Y] = \sum_{k \in \text{Range of } Y} \frac{1}{k} P(Y=k)$

Range of X
 $= \{0, 1, 2, \dots\}$
 Range of Y
 $= \{1, 2, 3, \dots\}$

$= \sum_{k=1}^{\infty} \frac{1}{k} P(X+1=k)$

$= \sum_{k=1}^{\infty} \frac{1}{k} \cdot P(X=k-1)$

$= \sum_{k=1}^{\infty} \frac{1}{k} \cdot \frac{\lambda^{k-1} e^{-\lambda}}{(k-1)!}$

$= e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{k!} = \frac{e^{-\lambda}}{\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{k!}$

$= \frac{e^{-\lambda}}{\lambda} (e^{\lambda} - 1) = \frac{1 - e^{-\lambda}}{\lambda} \text{ Ans}$